

Academic Validation Request

VALO V5.0 — Formal Verification Review

Version: V1.2

Date: May 15, 2026

Classification: Confidential — NDA Required

To:

Professor Leander Nikolaus Jehl and Professor Hein Meling University of Stavanger (UiS)

From:

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1. Purpose

VALO requests independent academic validation of the VALO V5.0 TLA+ formal specification. Specification is complete, implementation is built, security audit is passed. We need expert review to confirm correctness and produce a peer-reviewed validation report.

2. Architecture

Layer	Component	Language	Function
L1	Guardian	Rust	Safety decision gate — ACCEPT/REJECT in 43ns. 0 unsafe blocks. Only formally specified layer.
L2	Orchestrator	Python	Logging, routing, authentication, SHA-256 WORM audit log. Cannot override L1.
L3	Context Engine	Rust/Python	Source reliability tracking, distrust escalation (L0 Trusted → L4 Total Halt).

Architectural guarantee: L2 and L3 cannot override L1's decision. If L1 says REJECT, the system halts. This is enforced architecturally, not by convention.

3. What We Are Asking

1. Run TLC Model Checker

Execute TLC on formal-verification/ValoStateMachine.tla. Confirm the four invariants hold. We have not run this — your execution is the verification.

2. Review Invariants

Assess whether the four invariants correctly capture the safety properties.

3. Code Review (L1 only)

Review l1-guardian/src/ against the TLA+ spec. L2 and L3 are out of scope.

4. Scope Assessment

State independently what this verification proves and what it does not.

4. Materials

Technical Whitepaper v3	Full spec, 3-layer arch, honest validation status
TLA+ Source	formal-verification/ValoStateMachine.tla + .cfg
L1 Rust Source	l1-guardian/src/ — complete, under NDA
Security Audit	May 2026 — 0 critical findings
Signed NDA	v5.1, Scrive/BankID, Doc-ID: 09222115557577865024 ✓

Target: July 31, 2026 (8 weeks). NDA signed — ready to start immediately.

5. What UiS Receives

- ✓ Co-authorship on the validation report
- ✓ Novel TLA+ case study in critical infrastructure AI
- ✓ Contribution to EU AI Act compliance methodology
- ✓ Acknowledgement in all regulatory submissions

GitHub Repository Structure

Repository: Valo-Research-Geoup/valo-v5-core (private)

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VALIDATION.md	Formal validation specification and scope
WHITEPAPER.md	Technical whitepaper (Markdown)
SOVEREIGNTY.md	IP sovereignty and confidentiality declaration
Dockerfile	Container build definition
simulate.py	Infrastructure simulation runner
simulate_vaig.py	VAIG (AI inference guard) simulation
/l1-guardian/	
src/main.rs	L1 Guardian entry point — Rust
src/lib.rs	Core safety logic — ACCEPT/REJECT gate
src/valo_frame.rs	64-byte cache-aligned ValoFrame struct
src/validation_logic.rs	Safety measurement logic
src/crc32c.rs	CRC32C integrity verification
src/server.rs	Socket server — UDS/TCP transport
/l2-orchestrator/	
ai_gateway.py	AI model interface and routing
src/auth.py	Authentication layer
src/worm_log.py	SHA-256 hash-chained WORM audit log
src/propagator.py	Frame propagation and transport
/l3-context/	
src/context_engine.rs	Source reliability engine — Rust
src/distrust.rs	Distrust escalation matrix (L0–L4)
src/feeds.rs	External data feed management
/formal-verification/	

<code>ValoStateMachine.tla</code>	Primary TLA+ specification — L1 state machine
<code>ValoStateMachine.cfg</code>	TLC model checker configuration
<code>valo_v5.tla</code>	Extended TLA+ specification
/docs/	
<code>Valo_Research_Group__WP_Valo_5_0_L OCKED.pdf</code>	Locked whitepaper PDF
/sidecar/	
<code>vaig.py</code>	VAIG — AI inference guard sidecar
<code>client.py</code>	Test client

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